

Using Python and DS9 to Reduce and Analyse Astronomical Images

Reducing Images of Neptune and Moon Hunting.

Introduction

CCD cameras are the modern way to capturing celestial objects. CCD stands for charge-coupled device. These cameras have a 2D lattice of light-sensitive electronic micro-cells, which are also known as pixels.¹ These pixels are what the image sensor is to a standard DSLR camera, the pixels build up a picture by storing an electron every time they interact with an incoming photon.

CCD cameras have their faults, and can produce images with artifacts present that are produced by malfunctions within the camera itself. One such fault is bias, this is caused by a constant voltage applied to the CCD detector, the pixel structure means that this bias is spread unevenly throughout the frame. This is solved by taking a zero second exposure to give a bias frame. Another fault is the dark current, which arises due to the heat of the camera. This temperature will displace electrons within the CCD, which will be picked up by other pixels and registered as a light source. Variations in the sensitivity of the pixels mean that the same light source is perceived to be brighter or darker by different pixels.

The techniques to correct these faults will be performed in a python code to obtain reduced images of Neptune and its moons.

Methods

To begin, I found the bias overscan region to be in the 597-640 pixel range of each row of the images. Then when looking at the flat images, there are many white dots similar to tv static, this feature is due to the varying sensitivities of the pixels. The science images appear to have dust artifacts.

The program begins by importing *NumPy*, *Pylab*, *fits* from *astropy.io* and *interpolation* from *scipy.ndimage*. All images and their corresponding headers were then read in using three for loops, one for each folder. The images were appended into empty lists, and then converted into an arrays. Within the for loops, the exposure time of each image was printed to screen, which allowed me to determine which dark observation

belonged to the flat images and which belonged to the science images.

The median flux of each row within the bias overscan region (597-640) was measured. Then the bias overscan region was clipped off by reshaping the image. The median bias of each row was then subtracted from every row of the reshaped image using a for loop within a for loop. The outer loop cycling through the images in the folder and the inner cycling through each row of the image, subtracting the median bias each time. This process was done twice, once for the science images and once for the flat images.

Then the dark correction was performed by subtracting the corresponding dark data from the science and flat images. This simple subtraction was made using two loops.

I then took a flat field correction of the images to correct for each pixel within the CCD having a different sensitivity. All the flat images were combined into a master flat image using the *np.median* function. Using the *flatten* function I collapsed the master flat image into a one dimensional array, then, using *np.median* I found the median pixel value of the flat field. I divided the master flat image by this median pixel value to get the normalised median of all the flat field exposures. Then using the *Pylab* functions, I plotted a histogram of the average pixel value (shown in figure 1). The now normalised master flat data was written to a fits file, and when looking at it, it is clear that all the pixels have the same sensitivity; a constant grey 'fog' overlays the image. The flat field correction of the science images was then performed by dividing the science images by the normalised master flat.

Then, using the *min* and *max* functions I found the minimum and maximum values of pixel intensities in the science images. I used the difference between them as the range for a histogram of pixel intensities. Using a for loop, the most frequent value of the pixel intensity of each image was then taken from the largest bin in the histogram. This pixel intensity was subtracted from every pixel in the image, taking away the sky's contribution to our image.

I then loaded three corrected images into DS9, and made note of the coordinates of 4 dim, and hence distant stars in each frame. These distant stars are fixed in the sky, so by measuring their change of coordinates

in each frame, the pixel shift required was found. This table is shown below in Table 1. The required shift is performed using the *interpolation* function.

Frame	Object 1	Object 2	Object 3	Object 4	Average shift
1	270,2381	461,846	197,487	462,296	0,0
2	273,2365	464,829	201,471	467,279	-4,17
3	276,2342	467,806	204,447	470,257	-7,39

Table 1. A table of the coordinates of four background stars in the three reduced images and the subsequent average shift of the pixel frame required.

Finally, using the blink feature on DS9, I was able to flick between the fully corrected images. After having changed the contrast, I was able to spot multiple objects moving around Neptune.

Results

The histogram of the average pixel sensitivity, used to perform the flat field correction is shown in figure 1.

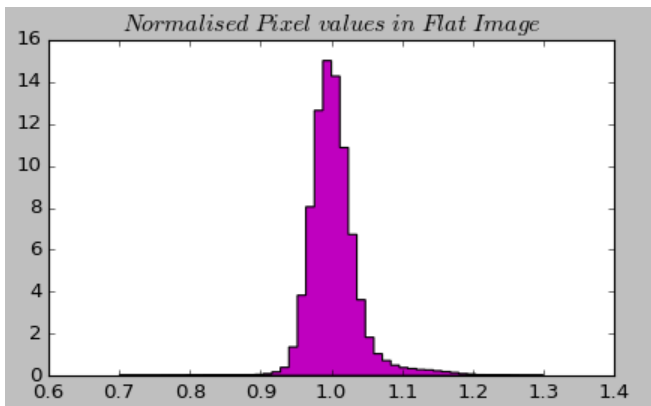
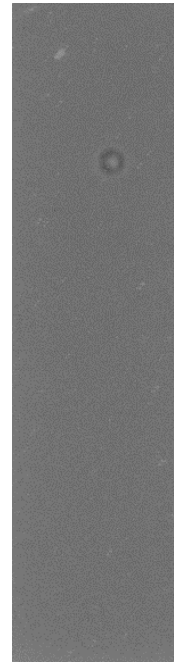


Figure 1. A histogram to show the average pixel sensitivity after the flat field normalisation. Where the x axis is the normalised pixel value and the y axis is the count.



The reduced and pixel shifted images obtained are displayed in figure 2.

Figure 2. The normalised, master flat image.

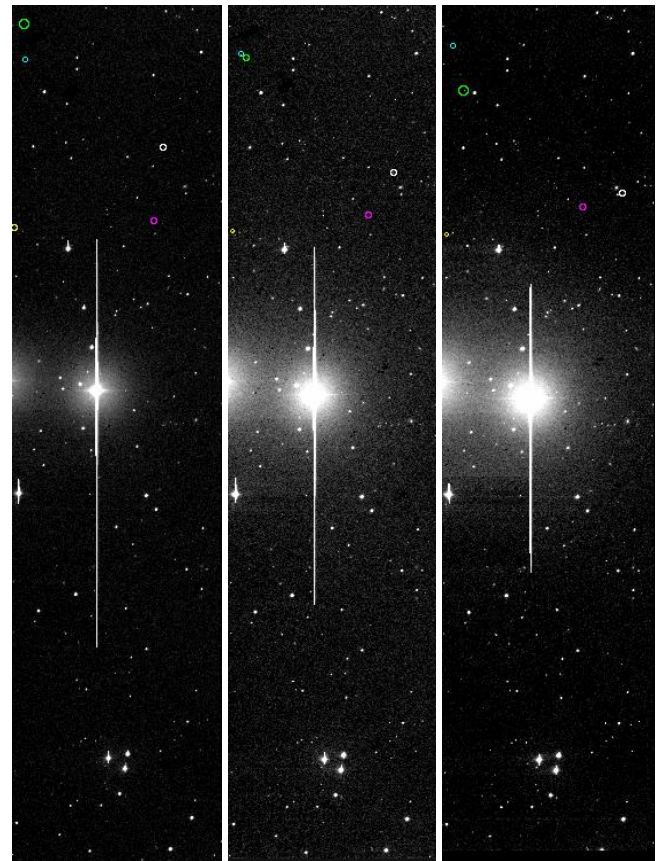


Figure 3. The three science images produced after all corrections have been performed. 5 moons have been highlighted with different coloured regions, all of them above Neptune.

The positions of Neptune are: (244, 1318), (249, 1308) and (251, 1303) for the first, second and third image respectively.

Discussion

The histogram in figure 1 shows that all the pixel sensitivities are equal to 1 within a ± 0.2 margin. The static white dots have been removed from the scientific images. In figure 2 you can clearly see that the pixel sensitivities have been set to equal 1. The circular smudge perhaps caused by dust is also observable in the master flat. This is a success since it is what is being subtracting from our scientific images. There is still the vertical white line bisecting Neptune, this is likely to be a saturation trail. This occurs when the CCD's pixels cannot handle the light from Neptune during the 180 second exposure. The pixels reached the maximum amount of electrons that they could store, and the electrons began filling up the neighbouring pixels.

The figures produced are clear enough to distinguish Neptune's moons from the background of stars, I highlighted 5 of these in different colours.

Conclusions

The python code written successfully reads in images of Neptune and performs bias, dark, flat fielding and sky subtraction corrections and aligns the images. The resulting images can be analysed using DS9 software to observe Neptune's Satellites, 5 were found and are detailed in a regions folder attached.

References

1. P. Martinez "A practical guide to CCD astronomy" Cambridge University Press (1998) pages: 1-6